

SHP ASSET MANAGER

LAZER EYE

User Manual — V2 Hardware



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SYSTEM OVERVIEW

The **Lazer Eye** is a LiDAR-based camera trigger. It continuously measures the distance to whatever is in front of it and fires your camera shutter the moment something enters a distance window you define – no button press required.

Common uses include wildlife photography (animal crosses the beam), action sports (athlete reaches a point), product photography, and long-exposure light painting.

WIRELESS RECEIVER OPTIONS

An optional wireless receiver sits at the camera end and receives trigger and wake signals from the Lazer Eye, eliminating a cable run between the sensor and your camera. Two V2 receiver types are supported:

RECEIVER		BEST FOR
LR Mini RX V2	v2	Compact receiver. Identifies itself to the app automatically once paired.
LR Full ESP32 RX V2	v2	Full-size receiver, same feature set as the Mini.

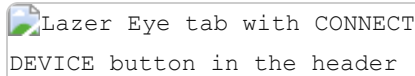
TRIGGER OUTPUT

The Lazer Eye has a wired trigger jack, and can optionally also trigger a paired V2 receiver wirelessly over ESP-NOW.

OUTPUT	WHAT IT DOES	BEST FOR
ESP-NOW wireless (optional)	Sends trigger and wake signals to a paired V2 receiver over 2.4 GHz. No cable required between the Lazer Eye and the camera.	Remote or awkward placements, water environments, moving rigs.

CONNECTING TO THE APP

The SHP Asset Manager app controls all Lazer Eye settings in real time. Connect via **Bluetooth** or **WiFi** – both give full control.



Tap ⚡ **CONNECT DEVICE** to start pairing

BLUETOOTH

- 1 Power on the Lazer Eye.
- 2 Ensure Bluetooth is enabled on your phone.
- 3 Open the app and tap the **LAZER EYE** tab at the bottom of the screen.
- 4 Tap ⚡ **CONNECT DEVICE** in the header and select your device from the scan LAZER_EYE_V2 list. The Lazer Eye appears as
- 5 Settings sync automatically. The dot in the header turns orange when connected.

The app remembers your device. On subsequent launches it reconnects silently if the Lazer Eye is in range.

WIFI

The Lazer Eye creates its own WiFi hotspot. WiFi is recommended when installing app or firmware updates – it is significantly faster than Bluetooth.

- 1 Power on the Lazer Eye.
- 2 On your LAZER_EYE network in OS WiFi settings. If you have set a Security PIN of 8 or more characters in the app, that PIN is also the join the hotspot password.
- 3 Return to the app. Tap ⚡ **CONNECT DEVICE** → *Connect via WiFi*.
- 4 Enter IP **192.168.4.1** and port **80**, then tap CONNECT.
- 5 Settings sync automatically once connected.

CONFIGURATION SETTINGS

The **CONFIGURATION** card is expanded by default at the top of the Lazer Eye tab. It holds the core trigger-zone settings, laser alignment, the LED toggles, and Wake Trigger.



Lazer Eye tab showing the configuration card with Min Distance, Max Distance, Shot Interval and Shots Per Trigger

Configuration card – Min/Max Distance, Shot Interval, Shots Per Trigger

MIN DISTANCE

0 cm up to Max Distance

The closest range at which the Lazer Eye will trigger. Objects nearer than this are ignored entirely.

MAX DISTANCE

Min Distance up to 1000 cm

The furthest range that triggers a shot. Objects beyond this are ignored. The orange zone on the distance bar shows your active trigger window.

SHOT INTERVAL

0.0 s to 30 s

Minimum time between successive triggers. Set to **0.0 s** to fire on every sensor reading. Increase to limit how rapidly the camera fires when a subject lingers in the zone.

DELAY BEFORE SHOT

0 s to 5 s

Pauses this long between the detection event and the shutter firing. Use for subjects that are still approaching the ideal frame position when detected. 0 = instant.

SHOTS PER TRIGGER

1 to 10

How many photos are taken each time the trigger fires. **1** for single capture; increase for burst sequences.

LASER ALIGNMENT

On / off – auto-off after 5 min

Fires the built-in alignment laser so you can see exactly where the sensor is aimed. Turns off automatically after 5 minutes, or tap the toggle again to stop it early.

STATUS LED

On / off

Blinks during each camera trigger and wake pulse. Disable in environments where stray light would disturb the shot.

POWER LED

On / off

Solid indicator that the Lazer Eye is powered on. Disable for stealth shooting or to minimise light pollution during long exposures.

All values are written to the device immediately when changed and are retained after power-off.

Do not stare directly into the alignment laser beam.

READING THE DISTANCE MONITOR

- > The large number at the top shows the live distance reading.
- > The **orange zone** on the bar marks your Min-Max trigger window. When the moving dot enters it the shutter fires and the display flashes **TRIGGER**.
- > Tap **CM/M** or **FT/IN** to switch the display unit.

WAKE TRIGGER

Many cameras enter a sleep or standby mode after a period of inactivity and miss the first shutter trigger after a long wait. Wake Trigger sends a short periodic pulse to keep your camera awake between triggers.

- 1 Scroll to the **WAKE TRIGGER** row inside the Configuration card.
- 2 Toggle the switch **ON** . The label turns orange when active.
- 3 Select a preset interval or enter a custom value.

PRESET	INTERVAL	RECOMMENDED FOR
30s	30 seconds	Cameras with very aggressive sleep settings
1 min	60 seconds	General use – most cameras
2 min	120 seconds	Cameras with 1-2 min auto-off
5 min	300 seconds	Long waits – wildlife setups
10 min	600 seconds	Very long waits
30 min	1800 seconds	Overnight setups
CUSTOM	1-1800 s	Any specific interval

When connected, a **NEXT PULSE IN** countdown appears at the bottom of the card showing when the next wake pulse will fire.

Set the wake interval slightly shorter than your camera's auto-off time. If the camera sleeps after 1 minute, use a 45-second interval.

ESP-NOW WIRELESS RECEIVER & PAIRING

The ESP-NOW WIRELESS RECEIVER card lets the Lazer Eye trigger a paired V2 receiver wirelessly, sending trigger and wake signals over 2.4 GHz instead of a cable run to the camera.

PAIRING A RECEIVER

- 1 Power on the wireless receiver and connect the Lazer Eye to the app (Bluetooth or WiFi).
- 2 In the Lazer Eye tab, find **ESP-NOW WIRELESS RECEIVER** card and turn its **ON** toggle.
- 3 Expand the card. Tap the **refresh** button next to **RECEIVER MAC ADDRESS** to read back the Lazer Eye's own wireless ID.
- 4 Find the MAC address printed on the receiver itself.
- 5 Enter it into the **PAIR TRANSMITTER MAC** field in the **AA:BB:CC:DD:EE:FF** format, then tap **PAIR TRANSMITTER**.
- 6 The receiver locks onto the first packet it gets and pairs automatically. The badge at the top of the card switches from **NOT PAIRED** to **PAIRED**.

Pairing is remembered by both devices. You do not need to re-pair after power-cycling either the Lazer Eye or the receiver.

WHAT APPEARS AFTER PAIRING

- > **RECEIVER TYPE** — shows MINI RX V2 or FULL ESP32 RX V2 once the receiver identifies itself.
- > **RX FIRMWARE** — the receiver's firmware version, once reported.
- > **TX battery icon** — shows the receiver's battery level, updated whenever it reports a reading wirelessly.

RESETTING A RECEIVER'S PAIRING

To pair a receiver to a different Lazer Eye, clear its stored pairing first:

- 1 Power off the receiver.
- 2 Hold its **BOOT button** while applying power, and keep holding for **3 seconds**, then release.

3

The receiver clears its stored pairing and waits for a new first packet from whichever Lazer Eye pairs to it next.

TX & RX OUTPUT MODES

The **TX OUTPUT** card controls the Lazer Eye's own wake signal and autofocus behaviour. Once a receiver has been paired at least once, an **RX OUTPUT** card appears below it with the same controls for the receiver.

WAKE SIGNAL

Sets the signal level used on the wake/focus output.

MODE	USE WHEN
Dry Contact	Default. Works with most camera wake/focus inputs.
HOT 3.3V	Only enable if your camera requires a positive voltage pulse on the focus pin rather than a ground closure. Most cameras (Sony multi-terminal, Canon N3, Nikon MC-DC2 compatible) use ground closure – check your camera's remote/shutter pinout before switching this on.

The app shows a confirmation warning the first time you switch either Wake Signal to HOT 3.3V, since it applies voltage instead of a simple contact closure.

AUTOFOCUS PRE-PULSE

Holds the wake/focus pin for a set time *before* the shutter fires, giving your camera's autofocus a head start on moving subjects.

- > Toggle **AUTOFOCUS PRE-PULSE** (TX OUTPUT) or **RX AUTOFOCUS PRE-PULSE** (RX OUTPUT) on.
- > Set **FOCUS HOLD TIME** (TX: 50-2000 ms) or **RX FOCUS HOLD TIME** (RX: 50-5000 ms) to how long your camera needs to lock focus – typically 200-600 ms.
- > The wake pin is held for that duration, then the shutter fires.

RX OUTPUT LEDS

The RX OUTPUT card also has its own **STATUS LED** and **POWER LED** toggles for the receiver, independent of the Lazer Eye's own LED settings in the Configuration card.

RX OUTPUT is greyed out until ESP-NOW is enabled and a receiver has been paired at least once. After that it stays editable even if the Lazer Eye is disconnected.

If a red • **Command not confirmed – tap to retry** banner appears after changing a Wake Signal mode, the Lazer Eye didn't confirm the change in time and the toggle reverted. Tap the banner to resend, or move closer to the device / switch from Bluetooth to WiFi.

ADVANCED SETTINGS

Tap the **ADVANCED** row to expand these controls. Most users can leave them at their defaults.



Full Lazer Eye tab showing the
ADVANCED row collapsed at the
bottom of the screen

ADVANCED row – tap to expand

SCAN RATE

10–500 Hz, default 100 Hz

How many distance readings per second the sensor takes. Higher values make the system more responsive to fast-moving subjects but draw more power.

PULSE SPEED

50–2000 ms, default 300 ms

How long the trigger output is held active per shot. Increase if the camera misses short triggers.

SMOOTHING

1–10

Averages recent readings before evaluating the trigger. Higher values smooth out readings over longer distances and reduce false triggers, at the cost of slightly more latency.

FLUX GATE

Off, or 5–200

Rejects low-confidence readings (weak signal returns), which filters phantom distance spikes from fast-moving subjects crossing the beam edge or reflective surfaces. Increase if you see false triggers; lower it or turn it off if the sensor stops detecting low-reflectivity subjects (e.g. dark clothing).

STATUS LEDS & BATTERY

LAZER EYE LEDS

LED	BEHAVIOUR	MEANING
Status LED	Single blink	A trigger or wake pulse just fired. Toggle in Configuration → Status LED.
Power LED	Solid on	Device is powered on and running. Toggle in Configuration → Power LED.

V2 RECEIVER LEDS

Both the LR Mini RX V2 and LR Full ESP32 RX V2 use the same LED behaviour, controlled independently from the Lazer Eye's own LEDs via the RX OUTPUT card's Status LED and Power LED toggles.

BATTERY

The Lazer Eye's battery percentage appears as a battery icon in the app header when connected. Once a receiver is paired and reporting, its battery level appears next to the **TX** battery icon in the ESP-NOW card.

Charge devices via their USB-C ports using any standard 5 V USB charger.

TROUBLESHOOTING

False triggers when nothing is in frame

Increase the **Flux Gate** value in Advanced – start at 20 and increase until false triggers stop. The sensor is catching low-confidence reflections from a background or shiny surface.

Sensor stops detecting subjects

Lower the **Flux Gate** value, or turn it off. It may be set too strict for dark or matte subjects (e.g. black clothing). Also confirm the Min-Max trigger zone covers the subject's position using the live distance monitor.

Camera misses the first shot after a long wait

Enable **Wake Trigger** at an interval slightly shorter than your camera's auto-off time. If the camera sleeps after 1 minute, use a 45-second interval.

Camera fires too late or too early relative to the subject

Adjust **Delay Before Shot** in Configuration to fine-tune the pause between detection and the shutter firing.

Camera misses short triggers

Increase **Pulse Speed** in Advanced. Some camera bodies require a longer signal to reliably register a shutter command.

Camera doesn't focus in time before the shot

Enable **Autofocus Pre-Pulse** in TX OUTPUT (or RX OUTPUT) and increase the Focus Hold Time until the camera locks focus reliably before the shutter fires.

A red "Command not confirmed" banner appears after changing a Wake Signal mode

The command was sent but the device did not confirm it in time – the toggle reverted to the last known good state. Tap the banner to resend. If it reappears repeatedly, move closer to the device or switch from Bluetooth to WiFi.

Receiver type or firmware shows blank after pairing

Power-cycle the receiver. On boot it reports its type and firmware version to the Lazer Eye; the app displays the correct values within a few seconds of the receiver coming online.

ESP-NOW shows NOT PAIRED

Make sure the receiver is powered on and the ESP-NOW toggle is on. Re-enter the receiver's MAC address in PAIR TRANSMITTER MAC and tap PAIR TRANSMITTER again. If it was previously paired to a different Lazer Eye, reset its pairing first (see Section 05).

Can't connect via Bluetooth

Toggle Bluetooth off and on and power-cycle the Lazer Eye. On iOS: Settings → Privacy → Bluetooth → confirm the SHP Asset Manager app has permission. The Lazer Eye appears as `LASER_EYE_V2` in the scan list.

An update is slow or failing

Switch to a WiFi connection – updates over WiFi are dramatically faster than Bluetooth. Settings → Hardware Settings → Check For Updates.

UPDATING FIRMWARE

- 1 Connect to the Lazer Eye in the app.
 - 2 Tap the  **gear icon** in the header to open Settings.
 - 3 Scroll to **Hardware Settings** and tap **CHECK FOR UPDATES** .
 - 4 If an update is available, tap **INSTALL UPDATE** . The device reboots and the app reconnects automatically.
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